ASSIGNMENT-5.4

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Task 1: Ethical Handling of User Data (Privacy & Anonymization)

Thonny - <untitled> @ 22 : 1

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```python
# Program to collect basic user data
# NOTE: Sensitive data should always be protected or anonymized

import hashlib

name = input("Enter your name: ")
age = int(input("Enter your age: "))
email = input("Enter your email: ")

# Hashing email to protect user identity
hashed_email = hashlib.sha256(email.encode()).hexdigest()

print("\n--- User Data (Protected) ---")
print("Name:", name)
print("Age:", age)
print("Email (Hashed):", hashed_email)

# Ethical Note:
# - Emails should never be stored in plain text
# - Hashing helps anonymize sensitive information
# - Personal data should be stored securely or avoided if not required
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
Enter your name: ROHAN
Enter your age: 20
Enter your email: ROHANNAMUTHABAJI@GMAIL.COM

--- User Data (Protected) ---
Name: ROHAN
Age: 20
Email (Hashed): 7b86b08ec0b7222a0930d88b0f8b6a16ac73734e48555921c88b3cc14a57a480
>>>
```

Task 2: Sentiment Analysis with Bias Awareness

Thonny - <untitled> @ 26 : 1

File Edit View Run Tools Help

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```python
# Simple sentiment analysis with bias awareness

def analyze_sentiment(text):
    positive_words = ["good", "happy", "excellent", "great"]
    negative_words = ["bad", "sad", "terrible", "poor"]

    score = 0
    words = text.lower().split()

    for word in words:
        if word in positive_words:
            score += 1
        elif word in negative_words:
            score -= 1

    return "Positive" if score > 0 else "Negative" if score < 0 else "Neutral"


# Ethical considerations:
# - Avoid biased or offensive words in training data
# - Use balanced datasets
# - Regularly audit sentiment outputs for fairness

print(analyze_sentiment("This product is excellent and good"))
print(analyze_sentiment("This service is bad and terrible"))
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
 Positive
 Negative
>>>
```

Task 3: Ethical Product Recommendation System

Thonny - <untitled> @ 31 : 1

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```python

    for product in products:
        # Avoid favoritism by recommending based on relevance only
        if product["category"] in user_history:
            recommendations.append(product["name"])

    return recommendations


# Example data
user_history = ["electronics", "books"]

products = [
    {"name": "Laptop", "category": "electronics"},
    {"name": "Novel", "category": "books"},
    {"name": "Shoes", "category": "fashion"}
]

recommended = recommend_products(user_history, products)

print("Recommended Products:", recommended)

# Ethical Notes:
# - Recommendations should be transparent
# - Avoid unfair bias toward sponsored or preferred products
# - Allow user feedback to improve fairness

```

Shell

```
>>> %Run -c $EDITOR_CONTENT
 Recommended Products: ['Laptop', 'Novel']
>>>
```

Task 4: Ethical Logging (Avoiding Sensitive Data)

Thonny - <untitled> @ 23 : 1

File Edit View Run Tools Help

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```python
# Logging functionality with ethical considerations

import logging

logging.basicConfig(
    filename="app.log",
    level=logging.INFO,
    format="%(asctime)s - %(levelname)s - %(message)s"
)

def login_user(username):
    # Do NOT log passwords or sensitive personal data
    logging.info(f"User login attempt: {username}")
    print("Login successful")


login_user("student_user")

# Ethical Logging Guidelines:
# - Never log passwords or emails
# - Logs should be minimal and purpose-driven
# - Protect log files from unauthorized access
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
  Login successful
>>>
```

Task 5: Responsible Machine Learning Model Usage

Thonny - <untitled> @ 25 : 1

File Edit View Run Tools Help

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```python
# Simple ML model example with responsible usage documentation

from sklearn.linear_model import LinearRegression
import numpy as np

# Sample training data
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 6, 8, 10])

model = LinearRegression()
model.fit(X, y)

# Prediction
prediction = model.predict([[6]])
print("Predicted value:", prediction)

"""
Responsible Usage Notes:
- This model is trained on a very small dataset
- Predictions may not generalize to real-world data
- Always validate accuracy using proper test data
- Ensure training data is unbiased and representative
- Provide transparency when deploying ML models
"""
```